Bags, Vectors & Transformers

Transcribing Audio and Video · Bonus, Part 2

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A Lot of Our Text Starts as Speech



Interviews
and focus groups



Debates
parliament, councils



Podcasts
radio, talk shows



Video
YouTube, TikTok, TV news

Transcribing by hand takes several hours per hour of audio. **Automatic speech recognition** (ASR) now does a good first pass, in many languages, on your own computer.

Whisper (OpenAI, 2022) is the best-known open ASR model. It runs locally, so recordings never have to leave your machine.

How Whisper Works



A transformer like BERT, but an **encoder-decoder**. First trained on 680,000 hours of web audio in about 100 languages (later versions on more), so it copes with accents, noise and Dutch.

Long recordings are handled in consecutive 30-second windows, each conditioned on the text so far.

model	params	
`tiny`	39 M	quick look
`base`	74 M	CPU default
`small`	244 M	good on a Colab GPU
`medium`	769 M	slower
`large-v3`	1.55 B	most accurate
`turbo`	809 M	near `large`, much faster

Measuring Accuracy: the Word Error Rate

WER: the minimum number of words to insert, delete or substitute to turn the transcript into the reference, divided by the number of reference words. The word-level twin of the CER from Bonus 1.

Test recordings with a known transcript. The notebook generates speech with a synthesizer (espeak-ng), so it knows exactly what was said. With your own data: hand-transcribe a few minutes as the reference.

Normalization changes the score

reference: the government has not explained them very well

transcript: The government hasn't explained them very well.

comparison	WER
raw text	0.50
lowercase, no punctuation	0.25
Whisper's English normalizer	0.00

Whisper's normalizer also expands contractions and writes numbers as digits. Always report how you normalized.

Languages, Long Recordings and Video

Languages

· detects the language from the first 30 s, or set `language="nl"`

· accuracy follows the amount of training audio: English best, Dutch good

· `task="translate"` gives an English translation

Timestamps

· every **segment** has a start and end time

· `word_timestamps=True` times every word

· link quotes back to the recording; measure pauses and speaking rate

Video

· Whisper reads most video files directly, via ffmpeg

· or extract the audio first:

```
ffmpeg -i clip.mp4 -vn
-ac 1 -ar 16000 clip.wav
```

smaller to store and archive

When Whisper Makes Things Up

During silence, music or noise, Whisper sometimes writes **fluent sentences nobody said**, or repeats a phrase over and over. Because the output reads naturally, these errors are easy to miss.

Koenecke et al. (2024) found hallucinations in about 1% of transcriptions of real speech, some of them harmful (invented violence, made-up medical terms). Classic ASR makes errors too, but it doesn't invent whole sentences.

Defenses

- flag segments with a high `no_speech_prob` or low `avg_logprob`
- flag **repeated phrases**
- `condition_on_previous_text=False`
- `hallucination_silence_threshold` (with word timestamps)
- trim long silences and music first
- **spot-check against the recording**, always for quotes you use

Speakers, Privacy and a Workflow That Holds Up

Who said what? Whisper doesn't know. **Speaker diarization**: pyannote.audio, or WhisperX (Whisper + pyannote). For two-person interviews, assigning speakers by hand is often quicker.

Voices are personal data. Get consent for automatic transcription too. Colab runs on Google's servers: run confidential audio locally.

Workflow

1. Try two or three model sizes on a few minutes of *your* audio, against a hand-made transcript
2. Transcribe everything with the chosen model; keep the timestamps
3. Flag suspicious segments; check them and a random sample against the audio
4. Correct the passages you quote
5. Report the model, settings and error rate

Faster: faster-whisper (same models, several times faster), or a Hugging Face `pipeline` as on Day 3.

In the Notebook

- Generating English and Dutch test recordings with a known transcript
- Transcribing with Whisper, with segments, confidence and word timestamps
- The word error rate, normalization, and model sizes compared
- Language detection, Dutch, and transcribe vs. translate
- Long recordings and video files
- Provoking a hallucination and flagging suspicious segments
- A folder of recordings to one dataset, with the real-time factor

`bonus/notebooks/2_transcribing_audio_and_video_exercises.ipynb` · 5 exercises · GPU recommended